HAOMING MENG

Toronto, Canada ⋄ 416-505-3736 ⋄ haomingmeng@gmail.com ⋄ Google Scholar
Canadian Citizen; U.S. TN-eligible

EDUCATION

University of Toronto (St. George Campus) September 2024 - May 2028 (Expected)
PhD in Mathematics. Advisor: Vardan Papyan.

· Select Coursework: Algebra, Topics in Geometric Analysis: Optimal Transport and its Applications, Variational Methods in Imaging and Neural Networks.

University of Toronto (St. George Campus) September 2023 - September 2024
MSc in Mathematics. Advisor: Vardan Papyan.

· Select Coursework: Mathematical Probability I & II (graduate), Optimization (graduate).

University of Toronto (St. George Campus) September 2019 - June 2023
Honours BSc in Mathematics (Specialist), Computer Science (Major), Statistics (Minor)

· Cumulative GPA: 3.95/4.00.
· Select Coursework: Advanced Real Analysis (graduate), Algebraic Topology (graduate), Riemannian Geometry (graduate), Nonlinear Optimization, Real Analysis, Complex Analysis, Differential Geometry, Groups Rings & Fields, Topology, PDEs, Deep Learning (graduate), Time Series Analysis.

PUBLICATIONS AND TECHNICAL REPORTS

(* indicates equal contribution.)

Gradient Smoothing: Coupling Layer-wise Updates for Improved Optimization.
Haoming Meng*, Anton Sugolov*, and Vardan Papyan.
International Conference on Machine Learning (ICML), 2026.

Sparse but Critical: A Token-Level Analysis of Distributional Shifts in RLVR Fine-Tuning of LLMs.
Haoming Meng, Kexin Huang, Shaohang Wei, Chiyu Ma, Shuo Yang, Xue Wang, Guoyin Wang, Bolin Ding, Jingren Zhou.
International Conference on Learning Representations (ICLR), 2026.

Transformer Block Coupling and its Correlation with Generalization in LLMs.
Murdock Aubry*, **Haoming Meng***, Anton Sugolov*, and Vardan Papyan.
International Conference on Learning Representations (ICLR), 2025.

Pushing Boundaries: Mixup's Influence on Neural Collapse.
Quinn Fisher*, **Haoming Meng***, and Vardan Papyan.
International Conference on Learning Representations (ICLR), 2024.

FIPO: Eliciting Deep Reasoning with Future-KL Influenced Policy Optimization.
Chiyu Ma, Shuo Yang, Kexin Huang, Jinda Lu, **Haoming Meng**, Shangshang Wang, Bolin Ding, Soroush Vosoughi, Guoyin Wang, Jingren Zhou.
Qwen Pilot Technical Report; Conference on Language Modeling (COLM), 2026.

Beyond Magnitude: Leveraging Direction of RLVR Updates for LLM Reasoning.
Kexin Huang, **Haoming Meng**, Junkang Wu, Jinda Lu, Chiyu Ma, Ziqian Chen, Xue Wang, Bolin Ding, Jiancan Wu, Xiang Wang, Xiangnan He, Guoyin Wang, Jingren Zhou.
International Conference on Learning Representations (ICLR), 2026.

Clipping Bottleneck: Stabilizing RLVR via Stochastic Recovery of Near-Boundary Signals.
Shuo Yang, Jinda Lu, Chiyu Ma, Kexin Huang, **Haoming Meng**, Qihui Zhang, Yuyang Liu, Bolin Ding, Guoyin Wang, Li Yuan, Jingren Zhou.
International Conference on Machine Learning (ICML), 2026.

The Heel of RLVR: Benchmark Glory Should Not Outpace Honest Measurement.
Shuo Yang, Chiyu Ma, Kexin Huang, Jinda Lu, Shaohang Wei, Xinpeng Liu, **Haoming Meng**, Yuyang Liu, Shangshang Wang, Minghao Zhu, Soroush Vosoughi, Guoyin Wang, Jingren Zhou, Li Yuan.
The Fortieth Annual Conference on Neural Information Processing Systems Position Paper Track (NeurIPS), 2026.

RESEARCH EXPERIENCE

University of Toronto & Vector Institute — Toronto, Canada
Graduate Researcher — *August 2023 - Present*

- Supervised by Prof. Vardan Papyan, focusing on advancing the empirical and theoretical understanding of deep learning.
- Developed Gradient Smoothing, an optimization method that couples parameter updates across depth to improve optimization, demonstrating its effectiveness across LLM pretraining and RL post-training, ViTs, and diffusion models (ICML 2026).
- Developed a Jacobian-based analysis of transformer block interactions, identifying layer-wise regularity and cross-layer singular vector coupling that emerges during training and correlates with generalization. (ICLR 2025).
- Investigated representation geometry under Mixup, using theoretical analysis and experiments to characterize its Neural Collapse structure (ICLR 2024).

Alibaba Group (Qwen Pilot) — Bellevue, Washington, USA
Research Intern — *June 2025 - September 2025*

- Conducted research on understanding and improving RL post-training for LLM reasoning as part of the Qwen Pilot team. Mentors: Guoyin Wang and Bolin Ding.
- Led an investigation into the mechanisms underlying RLVR reasoning improvements, designing experiments and developing large-scale pipelines for distributional analysis, RL training, and token-level interventions, revealing that a small subset of tokens is critical to reasoning gains (ICLR 2026).
- Contributed to the development and evaluation of novel RLVR methods, including improved token-level credit assignment, direction-aware policy updates, and techniques for stabilizing RL training via clipping (ICLR, ICML, COLM, NeurIPS 2026).

University of Toronto — Toronto, Canada
Undergraduate Researcher — *May 2022 - September 2022*

- Conducted research with Prof. Maryam Mehri Dehnavi on accelerating Neural ODE training, supported by an NSERC Undergraduate Student Research Award; investigated connections between optimal control and Neural ODEs, developing continuous optimization-inspired architectures to improve solver robustness and reduce adaptive step-size iterations.

Fields Institute for Research in Mathematical Sciences — Toronto, Canada
Undergraduate Researcher — *July 2021 - September 2021*

- Selected for the Fields Undergraduate Summer Research Program 2021 under Prof. Florian Shkurti on safe model-based reinforcement learning; worked on extending a model-free safe reinforcement learning method to the model-based setting by integrating a learned state-transition model.

TECHNICAL SKILLS

Programming Languages: Python, Java, R, C++

Libraries and Frameworks: PyTorch, Transformers, vLLM, verl, JAX, SciPy.

ACADEMIC HONORS AND AWARDS

NSERC Canada Graduate Research Scholarship—Doctoral	2025 - 2028
University of Toronto Faculty of Arts and Science Top Doctoral Fellowship	2024 - 2028
NSERC Undergraduate Student Research Award	2022
The Chancellor's Scholarship (University of Toronto Trinity College)	2022, 2021
University of Toronto Department of Computer Science Research Award	2021
Drew Thompson Scholarship (University of Toronto Trinity College)	2020
University of Toronto Scholars Program Award	2019
Governor General's Academic Medal (Secondary School Level)	2019

TEACHING EXPERIENCE

Mathematics Teaching Assistant Toronto, Canada
University of Toronto *September 2020 - Present*

- Worked as a Teaching Assistant for Calculus I (Fall 2020, Fall 2021, Fall 2023, Summer 2024, Fall 2024), Calculus II (Winter 2021, Winter 2024, Winter 2025), the Math Learning Centre (Fall 2021), Linear Algebra II (Winter 2023), ODEs (Winter 2026), Concepts in Abstract Mathematics (Fall 2025).